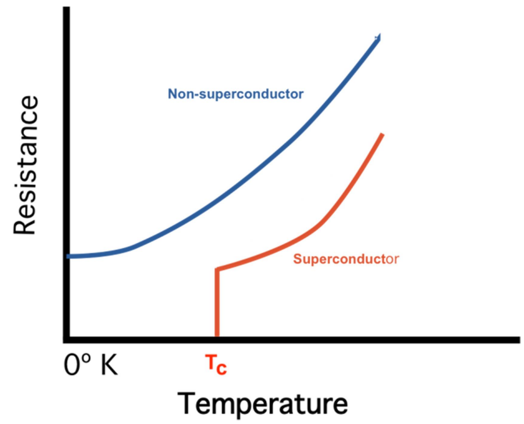
- Superconductors
- Properties of Superconductors,
- Meissner Effect,
- Type I and Type II Superconductors,
- BCS Theory (Qualitative),
- High-Tc Superconductors,
- Applications of Superconductors.

Introduction: Certain metals and alloys exhibit almost zero resistivity (i.e. infinite conductivity) when they are cooled to sufficiently low temperatures. This phenomenon is called superconductivity. This phenomenon was first observed by H.K. Onnes in 1911. He observed that the electrical resistivity of pure mercury drops suddenly to zero at about the boiling point of helium i.e., 4.2K. He concluded that mercury has passed into a new state which is called the superconducting state. The temperature at which the resistance disappears is called the transition temperature or critical temperature.



Definition of Superconductivity:

The electrical resistance of a substance suddenly drops to zero when it is cooled to sufficiently low temperature. This phenomenon is known as superconductivity. The material exhibits this phenomenon is known as superconductor. Examples: Hg, Nb, Zn, Zr, Cd, Al, etc.

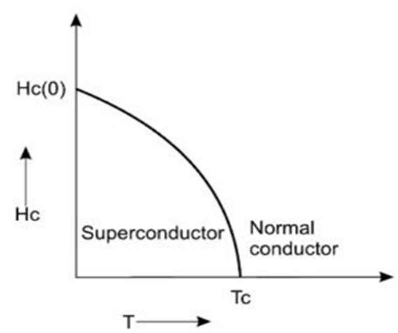
General Properties of Superconductors:

The temperature at which the transition from normal state to superconducting state takes place on cooling in the absence of magnetic field is called the critical temperature (Tc) or the transition temperature.

The following are the general properties of the superconductors:

- The transition temperature is different to different substances.
- ii. For a chemically pure and structurally perfect specimen, the superconducting transition is very sharp.
- The superconducting property of a superconductor will not lost by adding impurities to it, but the critical/transition temperature varies.
- The transition temperature (Tc) is different for different isotopes of superconductor. It is found to vary as the square root of atomic mass i.e., $T_c \propto M^{-1/2}$. This is known as isotope effect.
- At room temperature, superconducting materials have greater resistivity than other elements.
- Materials having high normal resistivity exhibit superconductivity.
- When a sufficient strong magnetic field is applied to a superconducting material below critical temperature its superconducting property is destroyed

- Superconductivity is found to occur in metallic elements in which the number of valence electrons lies between 2 & 8.
- There is no change in the crystal structure as revealed by X-ray Diffraction. This suggests that superconductivity is more concerned with the conduction electrons than with the atoms.
- Ferromagnetic and anti-ferromagnetic materials are not superconductors.
- The current in a superconducting ring persists for a very long time.
- Specific heat of superconductor varies exponentially with temperature.
- Thermal conductivity is usually lower in superconducting state.
- Entropy of a superconductor decrease rapidly on cooling below the transition temperature compared to normal state. This shows that superconducting state is more ordered than the normal state.
- **Effect of Magnetic Field:** Superconducting state of a metal depends on temperature and strength of the applied magnetic field. Superconducting property disappears if the temperature of the specimen is raised above Tc or a strong enough magnetic field is applied. At temperatures below Tc, in the absence of any magnetic field, the material is in superconducting state. When the strength of the magnetic field applied reaches a critical value then superconductivity disappears. The minimum value of applied magnetic field when the superconductor loses its superconductivity is called the critical magnetic field (Hc). At any given temperature below Tc there is a critical magnetic field Hc such that the superconducting property is destroyed on the application of magnetic field. The value of Hc decreases as the temperature increase.



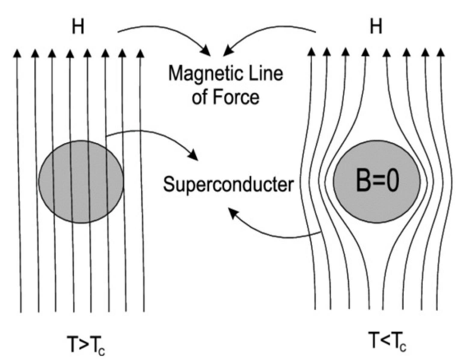
Variation of critical field with temperature

$$H_c(T) = H_0\,[1 - (T/T_c)^2]$$

Where H_0 is the critical field at 0K. H_0 and Tc are constants of the characteristics of the material

Meissner Effect:

When a weak magnetic field is applied to superconducting specimen at a temperature below transition temperature (Tc), then the magnetic flux lines are expelled from the specimen. Now the specimen acts as on ideal diamagnetic material. This effect is called Meissner effect. Meissner effect is reversible, i.e., when the temperature is raised from below Tc, at T = Tc the flux lines suddenly start penetrating and the specimen returns back to the normal state.



Under this condition, the magnetic induction (B) inside the specimen is given by $B = \mu_0(H + M)$ ---------------(1)

Where H is the external applied magnetic field and M is the magnetization produced inside the specimen. When the specimen is superconducting, according to Meissner effect inside the superconductor, B= 0. Hence,$_0$ (H + M) = 0 (or)M = - H

Thus the material is perfectly diamagnetic.

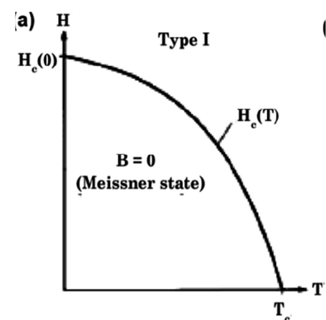
Magnetic susceptibility can be expressed as,

$$\chi = M/H = -1$$

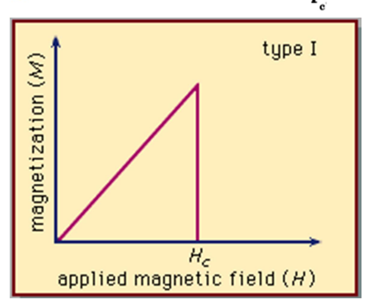
Types of superconductors:

Based on diamagnetic response Superconductors are divided into two types, i.e., type-I and type-II Superconductors.

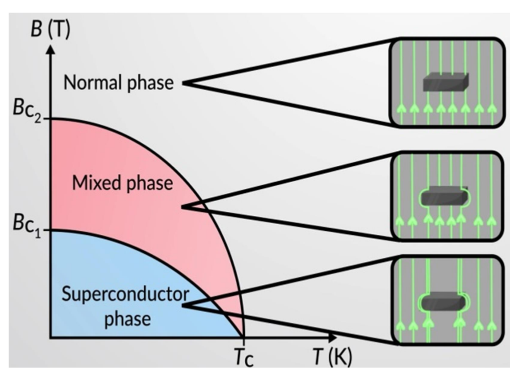
Type- I superconductors (or) Soft Super conductors:



- If the transition from superconducting state to normal state in the presence of magnetic field occurs sharply at the critical value H_c then the material is known as Type-I Superconductor.
- Superconductors exhibiting a complete Meissner effect are called type-1 superconductors.
- Type-1 superconductors also called Soft Superconductors.
- When the strength is gradually increased, at critical magnetic field (HC) the diamagnetism abruptly disappears.
- The critical field Hc is relatively low.
- No Mixed region or vortex region in this type of superconductors
- Materials with pure form are type I super conductors.
- Examples: - Al, Zn, Ga, Hg, and Sn etc.

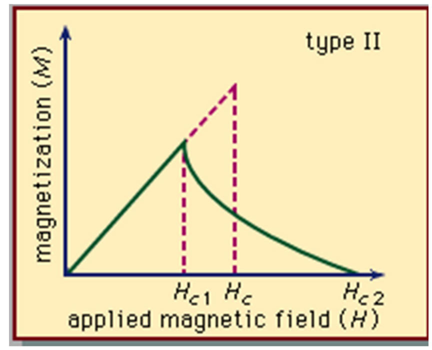


Type- II superconductors (or) Hard Super conductors:



- If the transition from superconducting state to normal state in the presence of magnetic field occurs gradually then the material is known as Type-II Superconductor.
- Type-II superconductors are also known as hard superconductors. They have high current densities.
- Materials with impurities or alloys are of type II superconductors.
- Type-II superconductor is characterized by two critical magnetic fields H_c1(called the lower critical field) and H_c2(called the upper critical field).
- For fields below Hc1, the superconductor expels the magnetic field from the body completely and behaves as a prefect diamagnetic. At HC1 the field begins to penetrate through the specimen. Penetration increases until HC2 is reached.

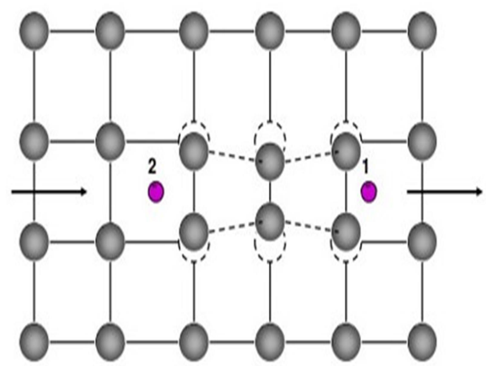
- Type-II superconductors show complete Meissner effect below Hc1 and allow the flux to penetrate the superconductor between Hc1 and Hc2. Between HC1 and HC2 the state of the material is called the mixed or vortex state. At HC2, the external field has completely penetrated into the body and destroyed the superconductivity. Now the material becomes normal state.
- Examples: - Zr, Nb, Lead-indium alloy, etc



BCS THEORY (EXPLANATION OF SUPERCONDUCTIVITY):

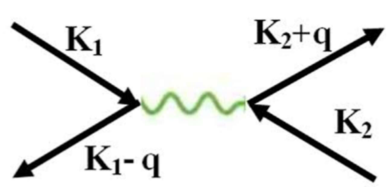
In 1957, Bardeen, Cooper and Schrieffer gave a theory to explain the phenomenon of superconductivity, which is known as BCS theory. The theory is based on the formation of Cooper Pairs.

In metals, the electrical resistance arises due to the collision of conduction (free electrons) with the vibrating ions of the lattice. In normal state, the force between the electrons is repulsive. In superconducting state, the force between two electrons becomes attractive due to formation of cooper pair.



When a current flows through a superconductor, and an electron (negative charge) comes near the positive ion core of the lattice, then the electron experiences an attractive force. Due to interaction between electron and ion core, the ion core is slightly displaced. This is known as lattice distortion. The distortion in the lattice then travels away as a mechanical wave (Phonons). Now, suppose that another electron comes near the distorted lattice. The lattice vibration (phonon) interacts with the second electron and hence there is a force of attraction between second electron and phonon. In this way, two electrons interact with each other through the lattice vibration. This process is called as electron – lattice-electron interaction via a phonon field (Mechanical waves).

When an electron with vector k_1, distorts the lattice and the lattice gains momentum. As a result, the momentum of the electron decreases. So a phonon of wave vector q is emitted. When another electron with wave vector k_2 absorbs the energy from phonon it gains momentum. Therefore, due to interaction, we have two electrons with wave vector k_1 - q and k_2 + q. The pair of electrons is called a Cooper pair. Therefore, Cooper pair is a bound pair of electrons formed by the interaction between the electrons in a phonon field. The two electrons which pair up have opposite momentum and spin.



The cooper pair of electrons moves on without suffering any deviation either by impurities or thermal vibrations. Hence, there is no exchange of energy between pair of electrons and lattice ions. If an electric field is established inside the substance, the electrons gain additional kinetic energy and give rise to a current. The main important point is that the pair of electrons does not transfer any energy to the lattice. Therefore, they do not get slowed down. As a consequence of this, the substance does not possess any electrical resistivity and the conduction is large. **When the pair of electrons flow in the form of Cooper pair, they do not encounter any scattering and the resistance factor vanishes**, i.e., conductivity becomes infinity which is named as superconductivity.

HIGH TEMPERATURE SUPERCONDUCTORS:

In 1986, Bednorz and Alex Muller in Zurich (Switzerland) discovered high temperature superconductivity in ceramics. They made a particular type of ceramic material which remained a superconductor at a temperature as high as 77 K. The high temperature superconductors are also called as high Tc materials.

Definition: Any Superconducting material with a transition temperature above 77K is in general called high Tc superconductor. All high temperature superconductors are different types of oxides of copper.

Important Observations:

1) All high temperature superconductors bear a particular type of crystal structure called the Perovskite structure.
2) The addition of extra copper oxygen layers into the structure unit of superconducting copper oxide complexes pushes the critical temperature to higher values.
3) The addition of any atom into copper oxide layer either brings down or destroys the effect of superconductivity.
4) The important observation is that the formation super currents in high Tc superconductors are direction dependent. The super currents are strong in copper- oxygen planes and weak in direction perpendicular to the planes.
5) In bulk materials, since they are ceramics, the flow of super currents has a restriction due to grain boundary effects. As a result, the critical current value is pushed down.

Few High Temperature Superconductors are given below:

a. YBa2Cu3O7 with Tc of 92K
b. Bi-Ca-Sr-Cu-O with Tc of 110K
c. Tl-Ba-Ca-Cu-O with a Tc of 120K
d. Hg-Ba-Ca-Cu-O with a Tc of 133 K
e. Hg-Ba-Ca-Cu-O with a Tc of 164 K (under pressure)
f. C60–Fullerences (2001) with a Tc of 110 K
g. InSnBa4Tm4Cu6O18 (2005) with a Tc of 150 K

Applications Of Superconductors:

1. Electric generators: Superconducting generators are very smaller in size and weight when compared with conventional generators. The low loss superconducting coil is rotated in an extremely strong magnetic field. Motors with very high powers could be constructed at very low voltage as low as 450V. This is the basis of new generation of energy saving power systems.

2. **Low loss transmission lines and transformers**: Since the resistance is almost zero at superconducting phase, the power loss during transmission is negligible. Hence electric cables are designed with superconducting wires. If superconductors are used for winding of a transformer, the power losses will be very small.

3. **Magnetic Levitation**: Diamagnetic property of a superconductor i.e., rejection of magnetic flux lines is the basis of magnetic levitation. A superconducting material can be suspended in air against the repulsive force from a permanent magnet. This magnetic levitation effect can be used for high speed transportation.

4. **Generation of high Magnetic fields:** Superconducting materials are used for producing very high magnetic fields of the order of50Tesla. To generate such a high field, power consumed is only 10kW whereas in conventional method for such a high field power generator consumption is about 3MW. Moreover in conventional method, cooling of copper solenoid by water circulation is required to avoid burning of coil due to Joule heating.

5. **Fast electrical switching**: A superconductor possesses two states, the superconducting and normal. The application of a magnetic field greater than Hc can initiate a change from superconducting to normal and removal of field reverses the process. This principle is applied in development of switching element cryotron. Using such superconducting elements, one can develop extremely fast large scale computers.

6. **Logic and storage function in computers**: They are used to perform logic and storage functions in computers. The current–voltage characteristics associated with Josephson junction are suitable for memory elements.

7. **SQUIDS (superconducting Quantum Interference Devices):** It is a double junction quantum interferometer. Two Josephson junctions mounted on a superconducting ring forms this interferometer. The SQUIDS are based on the flux quantization in a superconducting ring. Very minute magnetic signals are detected by these SQUID sensors. These are used to study tiny magnetic signals from the brain and heart. SQUID magnetometers are used to detect the paramagnetic response in the liver. This gives the information of iron held in the liver of the body accurately

8. **Magnetic resonance Imaging(MRI**) / NMR super conductinf magnets are used to create strong magnetic fields for high resolution medical imaging.

9. Sensitive devices for the measurement of Magnetic fields, electric current.

10. **Quantum computing:** super conducting circuits are employed in the development of quantum computers for their ability to maintain quantum states with minimum decoherence.

Properties of Super conductors (Short answer)

1. Zero Electric Resistance (Infinite Conductivity)
2. Meissner Effect: Expulsion of magnetic field
3. Critical Temperature/Transition Temperature
4. Critical Magnetic Field
5. Persistent Currents
6. Josephson Currents
7. Critical Current